

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Numerical

Let

$$A = \{n \in \mathbb{N} : n \text{ is a 3-digit number}\}$$

$$B = \{9k + 2 : k \in \mathbb{N}\} \text{ and}$$

$$C = \{9k + \ell : k \in \mathbb{N}\} \text{ for some } \ell (0 < \ell < 9)$$

If the sum of all the elements of the set $A \cap (B \cup C)$ is 274×400 , then ℓ is equal to

Q2: 24 Feb (Shift 2) - Numerical

The sum of first four terms of a geometric progression (G.P.) is $\frac{65}{12}$ and the sum of their respective reciprocals is $\frac{65}{18}$. If the product of first three terms of the G.P. is 1, and the third term is α , then 2α is

Q3: 25 Feb (Shift 1) - Single Correct

If $0 < \theta, \phi < \frac{\pi}{2}$, $x = \sum_{n=0}^{\infty} \cos^{2n} \theta$, $y = \sum_{n=0}^{\infty} \sin^{2n} \phi$ and

$z = \sum_{n=0}^{\infty} \cos^{2n} \theta \cdot \sin^{2n} \phi$ then

(1) $xyz = 4$

(2) $xy - z = (x + y)z$

(3) $xy + yz + zx = z$

(4) $xy + z = (x + y)z$

Q4: 25 Feb (Shift 1) - Numerical

Let $A_1, A_2, A_3, \dots$ be squares such that for each $n \geq 1$, the length of the side of A_n equals the length of diagonal of A_{n+1} . If the length of A_1 is 12 cm, then the smallest value of n for which area of A_n is less than one, is

Q5: 25 Feb (Shift 2) - Single Correct

The minimum value of $f(x) = a^{a^x} + a^{1-a^x}$, where $a, x \in \mathbb{R}$ and $a > 0$, is equal to:

(1) $a + \frac{1}{a}$

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(2) $a + 1$

(3) $2a$

(4) $2\sqrt{a}$

Q6: 26 Feb (Shift 1) - Single Correct

In an increasing, geometric series, the sum of the second and the sixth term is $\frac{25}{2}$ and the product of the third and fifth term is 25. Then, the sum of 4^{th} , 6^{th} and

8^{th} terms is equal to :

(1) 35

(2) 30

(3) 26

(4) 32

Q7: 26 Feb (Shift 1) - Single Correct

The sum of the infinite series $1 + \frac{2}{3} + \frac{7}{3^2} + \frac{12}{3^3} + \frac{17}{3^4} + \frac{22}{3^5} + \dots$ is equal to

(1) $\frac{9}{4}$

(2) $\frac{15}{4}$

(3) $\frac{13}{4}$

(4) $\frac{11}{4}$

Q8: 26 Feb (Shift 2) - Single Correct

The sum of the series $\sum_{n=1}^{\infty} \frac{n^2+6n+10}{(2n+1)!}$ is equal to :

(1) $\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

(2) $-\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

(3) $\frac{41}{8}e - \frac{19}{8}e^{-1} - 10$

(4) $\frac{41}{8}e + \frac{19}{8}e^{-1} + 10$

Q9: 26 Feb (Shift 2) - Numerical

If the arithmetic mean and geometric mean of the p^{th} and q^{th} terms of the sequence $-16, 8, -4, 2, \dots$ satisfy the equation $4x^2 - 9x + 5 = 0$, then $p + q$ is equal to

Answer Key**Q1** (5)**Q2** (3)**Q3** (4)**Q4** (9)**Q5** (4)**Q6** (1)**Q7** (3)**Q8** (3)**Q9** (10)

Q1 (5)

3 digit number of the form $9K + 2$ are $\{101, 109, \dots, 992\}$

$$\Rightarrow \text{Sum equal to } \frac{100}{2}(1093) = s_1 = 54650$$

$$274 \times 400 = s_1 + s_2$$

$$274 \times 400 = \frac{100}{2}[101 + 992] + s_2$$

$$274 \times 400 = 50 \times 1093 + s_2$$

$$s_2 = 109600 - 54650$$

$$s_2 = 54950$$

$$s_2 = 54950 = \frac{100}{2}[(99 + \ell) + (990 + \ell)]$$

$$1099 = 2\ell + 1089$$

$$\ell = 5$$

Q2 (3)

a, ar, ar^2, ar^3

$$a + ar + ar^2 + ar^3 = \frac{65}{12} \dots (1)$$

$$\frac{1}{a} + \frac{1}{ar} + \frac{1}{ar^2} + \frac{1}{ar^3} = \frac{65}{18}$$

$$\frac{1}{a} \left(\frac{r^3 + r^2 + r + 1}{r^3} \right) = \frac{65}{18} \dots (2)$$

$$\frac{(i)}{(ii)}, a^2 r^3 = \frac{18}{12} = \frac{3}{2}$$

$$a^3 r^3 = 1 \Rightarrow a \left(\frac{3}{2} \right) = 1 \Rightarrow a = \frac{2}{3}$$

$$\frac{4}{9} r^3 = \frac{3}{2} \Rightarrow r^3 = \frac{3^3}{2^3} \Rightarrow r = \frac{3}{2}$$

$$\alpha = ar^2 = \frac{2}{3} \cdot \left(\frac{3}{2} \right)^2 = \frac{3}{2}$$

$$2\alpha = 3$$

Q3 (4)

$$x = 1 + \cos^2 \theta + \dots \dots \dots \infty$$

$$x = \frac{1}{1 - \cos^2 \theta} = \frac{1}{\sin^2 \theta}$$

.....

$$y = 1 + \sin^2 \phi + \dots \dots \dots$$

Hints and Solutions

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$$y = \frac{1}{1 - \sin^2 \phi} = \frac{1}{\cos^2 \phi}$$

$$z = \frac{1}{1 - \cos^2 \theta \cdot \sin^2 \phi} = \frac{1}{1 - \left(1 - \frac{1}{x}\right)\left(1 - \frac{1}{y}\right)} = \frac{xy}{xy - (x-1)(y-1)}$$

$$xz + yz - z = xy$$

$$xy + z = (x + y)z$$

Q4 (9)



$$x = \frac{12}{\sqrt{2}} \quad y = \frac{12}{(\sqrt{2})^2}$$

∴ Side lengths are in G.P.

$$T_n = \frac{12}{(\sqrt{2})^{n-1}}$$

$$\therefore \text{Area} = \frac{144}{2^{n-1}} < 1 \Rightarrow 2^{n-1} > 144$$

Smallest $n = 9$

Q5 (4)

$$AM \geq GM$$

$$\frac{a^{ax} + \frac{a}{a^{ax}}}{2} \geq \left(a^{ax} \cdot \frac{a}{a^{ax}}\right)^{1/2} \Rightarrow a^{ax} + a^{1-ax} \geq 2\sqrt{a}$$

Q6 (1)

$$ar + ar^5 = \frac{25}{2}$$

$$ar^2 \times ar^4 = 25$$

$$a^2 r^6 = 25$$

$$ar^3 = 5$$

$$a = \frac{5}{r^3}$$

$$\frac{5r}{r^3} + \frac{5r^5}{r^3} = \frac{25}{2}$$

Hints and Solutions

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$$\frac{1}{r^2} + r^2 = \frac{5}{2}$$

$$\text{Put } r^2 = t$$

$$\frac{t^2+1}{t} = \frac{5}{2}$$

$$2t^2 - 5t + 2 = 0$$

$$2t^2 - 4t - t + 2 = 0$$

$$(2t - 1)(t - 2) = 0$$

$$t = \frac{1}{2}, 2 \Rightarrow r^2 = \frac{1}{2}, 2$$

$$r = \sqrt{2}$$

$$= ar^3 + ar^5 + ar^7$$

$$= ar^3 (1 + r^2 + r^4)$$

$$= 5[1 + 2 + 4] = 35$$

Q7 (3)

$$s = 1 + \frac{2}{3} + \frac{7}{3^2} + \frac{12}{3^3} + \frac{17}{3^4} + \frac{22}{3^5} + \dots$$

$$\frac{s}{3} = \frac{1}{3} + \frac{2}{3^2} + \frac{7}{3^3} + \dots \dots \infty$$

$$\frac{2s}{3} = 1 + \frac{1}{3} + \frac{5}{3^2} + \frac{5}{3^3} + \dots \dots \infty$$

$$\frac{2s}{3} = \frac{4}{3} + \frac{5}{3} \left\{ \frac{1/3}{1 - \frac{1}{3}} \right\} = \frac{5}{6} + \frac{4}{3} = \frac{13}{6}$$

$$s = \frac{13}{4}$$

Q8 (3)

$$\sum_{n=1}^{\infty} \frac{n^2+6n+10}{(2n+1)!}$$

$$\text{Put } 2n + 1 = r, \text{ where } r = 3, 5, 7, \dots$$

$$\Rightarrow n = \frac{r-1}{2}$$

$$\frac{n^2-6n+10}{(2n+1)!} = \frac{\left(\frac{r-1}{2}\right)^2 + 3r - 3 + 10}{r!} = \frac{r^2 + 10r + 29}{4r!}$$

$$\text{Now } \sum_{r=3,5,7} \frac{r(r-1)+11r+29}{4r!} = \frac{1}{4} \sum_{r=3,5,7,\dots} \left(\frac{1}{(r-2)!} + \frac{11}{(r-1)!} + \frac{29}{r!} \right)$$

$$= \frac{1}{4} \left\{ \left(\frac{1}{1!} + \frac{1}{3!} + \frac{1}{5!} + \dots \right) + 11 \left(\frac{1}{2!} + \frac{1}{4!} + \frac{1}{6!} + \dots \right) + 29 \left(\frac{1}{3!} + \frac{1}{5!} + \frac{1}{7!} + \dots \right) \right\}$$

$$= \frac{1}{4} \left\{ \frac{e - \frac{1}{e}}{2} + 11 \left(\frac{e + \frac{1}{e} - 2}{2} \right) + 29 \left(\frac{e - \frac{1}{e} - 2}{2} \right) \right\}$$

Hints and Solutions

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$$= \frac{1}{8} \left\{ e - \frac{1}{e} + 11e + \frac{11}{e} - 22 + 29e - \frac{29}{e} - 58 \right\}$$

$$= \frac{1}{8} \left\{ 41e - \frac{19}{e} - 80 \right\}$$

Q9 (10)

Given, $4x^2 - 9x + 5 = 0$

$$\Rightarrow (x - 1)(4x - 5) = 0$$

$$\Rightarrow A.M = \frac{5}{4}, G.M = 1 \quad (Q.A.M > G.M)$$

Again, for the series

$$-16, 8, -4, 2, \dots$$

$$p^{\text{th}} \text{ term } t_p = -16 \left(\frac{-1}{2} \right)^{p-1}$$

$$q^{\text{th}} \text{ term } t_q = -16 \left(\frac{-1}{2} \right)^{q-1}$$

$$\text{Now, } A.M = \frac{t_p + t_q}{2} = \frac{5}{4} \text{ \& } G.M = \sqrt{t_p t_q} = 1$$

$$\Rightarrow 16^2 \left(-\frac{1}{2} \right)^{p+q-2} = 1$$

$$\Rightarrow (-2)^8 = (-2)^{(p+q-2)}$$

$$\Rightarrow p + q = 10$$